

CANONICAL LESSON

# Satellites Need Ground Truth

Decision-centred foundations for accountable sector intelligence

|                 |                                   |
|-----------------|-----------------------------------|
| Production Unit | PU-B07-C15                        |
| Notebook        | V7_A_N10                          |
| Status          | Controlled educational draft v0.1 |
| Data boundary   | Synthetic instructional data only |

**Central decision:** Integrate remote-sensing evidence only after documenting spatial resolution, temporal alignment, cloud and sensor limits, ground truth, and uncertainty.

## Learning outcomes

### Why this matters

Satellite signals can be visually persuasive while measuring the wrong land cover, season, geography, or agricultural condition.

The analytical workflow is useful only when evidence, uncertainty, authority, and action are connected explicitly. The model supports professional judgement; it does not replace accountable institutional authority.

### Worked cross-sector example

A vegetation anomaly is checked against field observations and crop masks before it informs a production-risk brief.

**Boundary:** The example supports learning and review. It does not constitute an official statistic, eligibility determination, enforcement action, clinical judgement, or operational authorization.

## Practice and reflection

- 1.
- Stratify validation sampling.
- 2.
- Explain resolution limits.
- 3.

- Add a change-detection check.

## Answer key

- 1.
- Sample across zones, crop prevalence, terrain, and confidence classes with known inclusion probabilities.
- 2.
- Mixed pixels and small fields limit interpretation below pixel scale.
- 3.
- Hold legend and processing constant, quantify classification uncertainty, and verify apparent change in the field.

## Operational readiness checklist

- Purpose and decision owner are named.
- Evidence definitions, provenance, and lawful access are documented.
- Uncertainty and abstention are visible.
- Human review, escalation, monitoring, incident response, and retirement are assigned.
- Equity, accessibility, security, and sector-specific approval are complete.